

## Lesson 4.1: Identifying Solutions

**Location:** Classroom

**Grade Level:** High school

**Time:** 50 minutes

**Driving Question**

- How can we communicate solutions to climate change?

**Lesson Synopsis**

In this lesson, students get ready to inform the public about local climate change issues through a science communication product.

Students will work in groups to identify the message they want to communicate, drawing from the models they created and revised in previous modules. Students will be introduced to key principles of science communication and use GenAI tools to practice these principles.

**What students will figure out:**

There are key principles of science communication to make sure our messages reach the targeted audience.

- **Intentionality** is being deliberate about who we are communicating to.
- **Reciprocity** refers to valuing different perspectives, and
- **Reflexivity** means being reflective about our science communication approaches.

### LEARNING OUTCOMES

| By the end of this investigation, students will be able to...                                                     | You can assess this using...                                                                                                                           |
|-------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Explain what the science communication principles are and apply them to designing and communicating solutions. | Students' science communication plans that they created during the <i>Anchoring Phenomena</i> section.                                                 |
| 2. Identify perspectives and questions that GenAI can simulate and help to revise their solutions.                | Students' prompts to create the chatbot personas in the <i>Investigate</i> section.<br>Student-chatbot interactions in the <i>Investigate</i> section. |

**Building towards NGSS and AI Literacy:**

- **HS-ESS2-4:** Use a model to describe how variations in the flow of energy into and out of Earth's systems result in changes in climate.
- **AI Literacy's Big Idea #4: Natural Interaction.** Describe several approaches to natural language processing (e.g., how a chatbot can work using keywords, templates, or machine learning like GenAI). In this lesson, students will interact with several generative AI chatbots. They will develop prompts for the generative AI chatbots to test principles of science communication.

**LESSON OVERVIEW**

| Section                 | Time       | What Happens                                                                                                                                                                                        |
|-------------------------|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Anchoring Phenomena     | 10 minutes | Students brainstorm key components of science communication products.                                                                                                                               |
| Investigate             | 10 minutes | Students create a first draft of their science communication message, based on the information and models they have collected in previous modules.                                                  |
| Navigate                | 20 minutes | Students test their message with a chatbot, which roleplays as a specific audience group. Students learn about key science communication principles: <b>Intentionality</b> and <b>Reciprocity</b> . |
| Navigate                | 8 minutes  | Students explore the principle of <b>Reflexivity</b> as they reflect on how they might change their science communication message based on the feedback they gather from the chatbot.               |
| Putting Pieces Together | 2 minutes  | Students reflect on what else they might need to know to improve and develop their science communication product.                                                                                   |

**MATERIALS**

- [Lesson 4.1 slideshow](#)
- [Science journal](#)
- [AISciComm chatbot website](#)
- [Teacher Rubric](#) for science communication products (referenced throughout module 4)
  - [Student Rubric Checklist](#) (provided in science journal, p.60)

**BEFORE YOU START TEACHING**

1. Review [AISciComm chatbot website](#) to model and/or support students.

2. Review the Science Communication Product [Student Checklist](#) (also in Science Journal p. 60) and [Teacher Rubric](#) to know the expectations of students.
3. Throughout the program, students will be working in teams of three (preferably).
4. Make sure that you have a way to project a video with sound.
5. The science journal is intended to be a digital notebook. However, if you decide to print copies for students, they may need a writing instrument (pen, pencil, etc.).

### **WHERE WE ARE GOING AND NOT GOING**

**Where We Are Going:** This lesson is the beginning of Module 4, where students will create a science communication product that showcases climate change issues and how they affect Southern California's communities and environment. Students will be using the models and information they have gathered from the previous modules to make sure they include community perspectives, as well as local issues that may resonate with their audience. By the end of the lesson, students should understand how to construct effective science communication products and have a plan for how they will build their own.

Students will encounter the terms: **intentionality**, **reciprocity**, and **reflexivity** when planning the effectiveness of their science communication tool. Students will be encouraged to think about how their message can be conveyed in a meaningful and engaging manner.

**Where We Are NOT Going:** Students will be creating plans to make a science communication product that focuses on local climate change issues. They will not explore global climate change issues. Students will use GenAI tools (e.g., the AISciComm chatbots) to test the effectiveness of their messaging. They will not use these tools to construct the message, but rather to simulate how different audiences may perceive their messages.

**LEARNING SEQUENCE**

Lesson plan notes are also provided on the slide notes.

**ANCHORING PHENOMENON: Introducing Today's Lesson (10 minutes)**

1. Open the slideshow to **Slide 1** and introduce lesson 4.1.
2. Advance the slideshow to **Slide 2** to invite students to reflect on their experience so far and share as a group:
  - a. *What has been the most interesting experience in the curriculum so far?*
  - b. *What is something that you learned about that many others may not know, but should know?*

Then, tell students: We can communicate this important information by creating a message, which is the focus of Module 4.

3. Advance the slideshow to **Slide 3** and introduce the topic of using science communication products to convey a message about climate change to their local and respective communities.
4. Advance the slideshow to **Slide 4** and share the **Central Inquiry Question: What makes a science communication message clear, accurate, and engaging?** Explain that in the following slides, we will understand the characteristics of a strong science communication message and see examples to highlight parts we like and resonate with. We will also see other components that we could critique and suggest improvements.
5. Advance the slideshow to **Slide 5**. Share that inclusive and effective science communication often involves five key areas: clarity and language, content, knowledge organization, style, and audience engagement.
  - a. For this lesson, we are focusing on (1) **clarity and language** and (2) **content**.
    - i. Explain how **clarity and language** help us deliver complex information so that the public or our target audience can understand. This may often include translating or including language for diverse populations.
    - ii. Explain how **content** ensures that the information is not only accurate but also has connections to everyday life or your personal/community life.
6. Advance the slideshow to **Slide 6**. Share how this is an example of a science communication product created by high school students to advertise a community event focused on learning more about the ocean and its conservation.
  - a. Ask: *What is something you notice and wonder about?*
7. On **Slide 7**, reflect with students the **Central Inquiry Question: What makes a science communication message clear, accurate, and engaging?** Take a closer look at the science

communication product example's message, scientific accuracy, and real-world and personal connection.

- a. **Message:** What is the key message?
  - b. **Scientific accuracy:** How does it show scientific concepts?
  - c. **Real-world and personal connections (engagement of message)-** How does it connect science to local impacts, everyday life, and/or personal stories/ culture?
  - d. **Discuss:** What works well? And what can be improved?
8. Advance the slideshow to **Slide 8**. Based on the five components of inclusive and effective science communication, share with students that a [student checklist](#) has been created to guide their review and design of science communication products.
9. Remind students on **Slide 9** that our focus is thinking about the message we want to share. Therefore, we are focusing on the first three sections of the student checklist: language and clarity, science accuracy, and real-world and personal connections.
  - a. Explain that the checklist provides guiding questions for students to reflect on when reviewing and designing their products.
10. Advance the slideshow to **Slide 10**. Explain to students that the following slides will show four different possible products that students could create.
  - a. Instruct students to use the Student Science Communication Product checklist on p. 60-61 in the Science Journal to review and take notes on 2-3 different science communication products ([Slides 11-14](#)).
11. **Slide 11** shows an example social media post.
  - a. Ask students to use the student checklist to reflect if the **science communication message is clear, accurate, and engaging**.
12. Advance to the next several slides to show different examples of science communication products. **Slide 12:** a podcast; **Slide 13:** a board game; **Slide 14:** a town hall presentation.
  - a. With each slide, ask students the same question: *What makes a science communication message clear, accurate, and engaging?*
13. Advance to **Slide 15** for a whole-class discussion.
  - a. *As you look across all four products, which one immediately grabbed your attention, and what specific detail made it stand out before you even read the main message?*
  - b. *Pick one product that struggled to make its point clear. What is **one specific revision** you would suggest to improve its message?*
14. Move on to **Slide 16** and explain to students that these different science communication products share key components of their message. There are several questions to keep in mind when constructing them:
  - a. **Issue:** *What do you want people to know about climate change's effect on the ocean?*
  - b. **Action:** What is the action that people should take?

- c. **Connection:** *What do you want people to learn or feel as a result of the interaction?*
- d. **Data:** *What data or evidence supports your message? What should you include in the communication?*

15. Inform students that now that they have gathered information about how climate change will impact the things their community values about Southern California's ocean, it is time to start thinking about how to share this information. Over the next few lessons, they will work with their teams to construct a message to share with their community.

### **INVESTIGATE: Choosing your Message (10 minutes)**

1. Let students join their project team and advance to **Slide 17**. Have students pull up the computational models they have created and revised in Modules 2 and 3 and tell them that they will be using their models to help them identify what message they want to share with their community.
2. With their teams, have students identify ([Science Journal, Module 4.1: Identifying Solutions, Student Task: Communicating Solutions, p. 62](#)):
  - a. The **specific climate issue** that their group is addressing. Example student ideas might include urban heat islands, coastal erosion, rising sea level, etc. Guiding prompts for students: *If someone only remembers one thing from your model, what should it be?*
    - i. The issue can be tied to what students added to their models in Module 3.5.
    - ii. If interview notes (e.g., from Module 1) are needed, remind students to go back to their [science journal](#).
  - b. The **action** that people should take.
  - c. The **message** that they are communicating and what they want their audience to know or feel at the end. Guiding prompts: *Does your message raise awareness, change perception, or inspire action?*
  - d. The **data** or **evidence** supporting their message. The data are taken from previous modules.

### **NAVIGATE: Improve your Message with Science Communication Principles (20 minutes)**

1. Advance to **Slide 18**. Explain to students that we will now discuss science communication principles to make our message more effective. Have the whole class respond to the question: *What does effective science communication look like to convey your message to the right audience?*
  - a. The class brainstorms some adjectives that describe good science communication (e.g., clear, easy to understand, catchy, etc.)
2. Advance to **Slide 19**. Introduce to students the first science communication principle:

**Intentionality.** This principle answers the question: *Who is your target audience? How do you communicate with this specific audience?*

Possible student ideas might include the general public (e.g., peers, community members, social media audiences), decision-makers (e.g., local government officials, school board, city planners), stakeholders (e.g., farmers, fishermen, business owners, environmental groups), future generations, and other science communities.

**Guiding prompts:** *Would this message look different if you were telling it to a policymaker vs. your younger sibling? What do you want this audience to feel: concern, hope, urgency, or empowerment?*

3. Advance to **Slide 20**. Explain to students that effective science communication also involves **Reciprocity**. This principle answers the question: *How can you test your message? What are the audience's approaches to your opinions on the issues you are trying to communicate?*
4. Advance to **Slide 21**. Have students come back to their project team. Explain to students that they will practice the intentionality and reciprocity principles by selecting a target audience and testing the message they have drafted in the Investigate section.
  - a. The student teams will be working on the same computer.
  - b. The team will choose an audience (a chatbot) from the 5 chatbots they have been introduced to in Module 2. This chatbot will be the target audience for their selected messages.
5. Advance to **Slide 22**. Explain to students that they can select 1 of the predefined messages OR work with the messages they have created about their models.
6. Advance to **Slide 23**. Explain to students that there are two tasks that the student teams will perform in testing their messages with the selected chatbot.
  - a. The teams will chat with the chatbot to get to know the audience. Example chat message: "I want to communicate with a {**target audience**} (for example, fisherman) about {**the issue the group is targeting**}. What do you, as a {**target audience**}, know about this topic? How do you feel about this topic?"
  - b. The team will also chat with the chatbot to test their message. Example chat messages:
    1. "Here is my science communication message. What do you think about it?"
    2. "What is a better way to phrase this?"
    3. "Is there a metaphor or illustration that I could use for this message that you could relate to?"

**[Extension; OPTIONAL]** Large language models can get more precise and reveal their underlying reasoning process when we prompt them to "explain your thinking step by step". To get to know the audience, students can include in the chat messages for the chatbots to "explain your thinking step by step".

**NAVIGATE: Explore the Principle of Reflectivity (10 minutes)**

1. Advance to **Slide 24**. Introduce students to the third principle of science communication: **Reflectivity**. This principle answers the question: *How can you reflect on your science communication plan and the impact it has?*
2. Invite students to discuss in their project team and complete their [science journal entry on Module 4.1 Identifying Solutions \(Student Reflection\)](#), p. 65:
  - What did you learn about communicating your message effectively?
    - Consider the three science communication principles:
    - **Intentionality**: How did thinking about your audience shape your message?
    - **Reciprocity**: What did you learn from testing your message and listening to feedback?
    - **Reflectivity**: How did you reflect on your choices and improve your message?
  - How did feedback change your thinking and message? Explain what you revised and why it made the message stronger or clearer.

**PUTTING PIECES TOGETHER AS A CLASS: Share our Idea (5 minutes; OPTIONAL)**

1. Call on a few student teams to share their reactions to the science communication principles and how they might apply them to science communication.
2. Students share what else they might need to know so that they can be **intentional** about their audience, **reciprocal** in building on their audience's understanding, and **reflective** about their approach.